

3.3 Component 3 – Chemistry: Elements, mixtures and compounds

Matter is composed of tiny particles called atoms and there are about 100 naturally occurring types of atoms called elements.

Elements are shown in the periodic table and are either metals or non-metals. Atoms are the building blocks for all substances. When two or more elements combine chemically a compound is produced.

Different substances have different combinations of atoms joined together in different ways, which gives them different properties, such as whether they are solid, liquid or gaseous at room temperature. Many materials we use are mixtures. Mixtures can be separated by processes such as filtration. Polymers have many useful applications.